

- 3 Two progressive water waves X and Y travel along a straight line from point A to point B. The variation of displacement of the waves with distance from A at an instant in time is shown in Fig. 3.1.

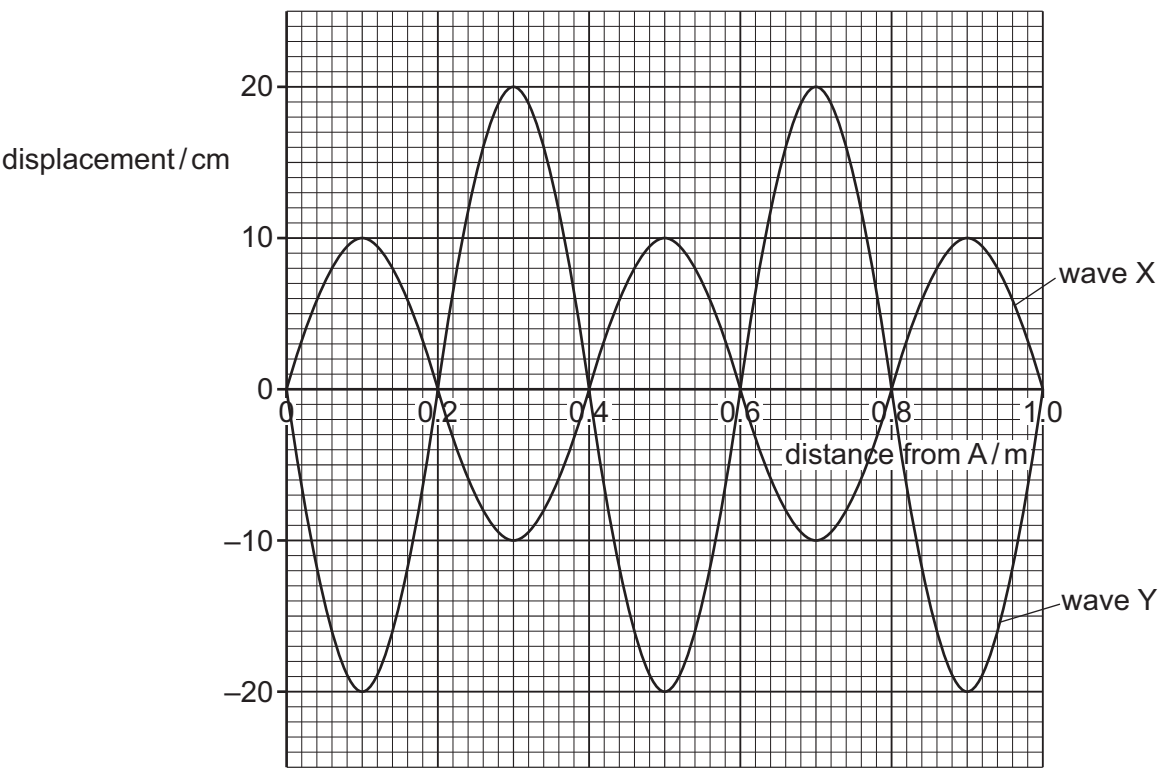


Fig. 3.1

- (a) State the amplitude of wave X. [1]
- (b) Both waves have frequency 16 Hz.
- (i) Determine the speed of wave X. [2]
- (ii) State and explain whether X and Y are coherent. [1]
- (c) Wave X and wave Y superpose to form a resultant wave.

On Fig. 3.2, sketch the variation of displacement of the resultant wave with distance from A at the instant of time shown in Fig. 3.1.

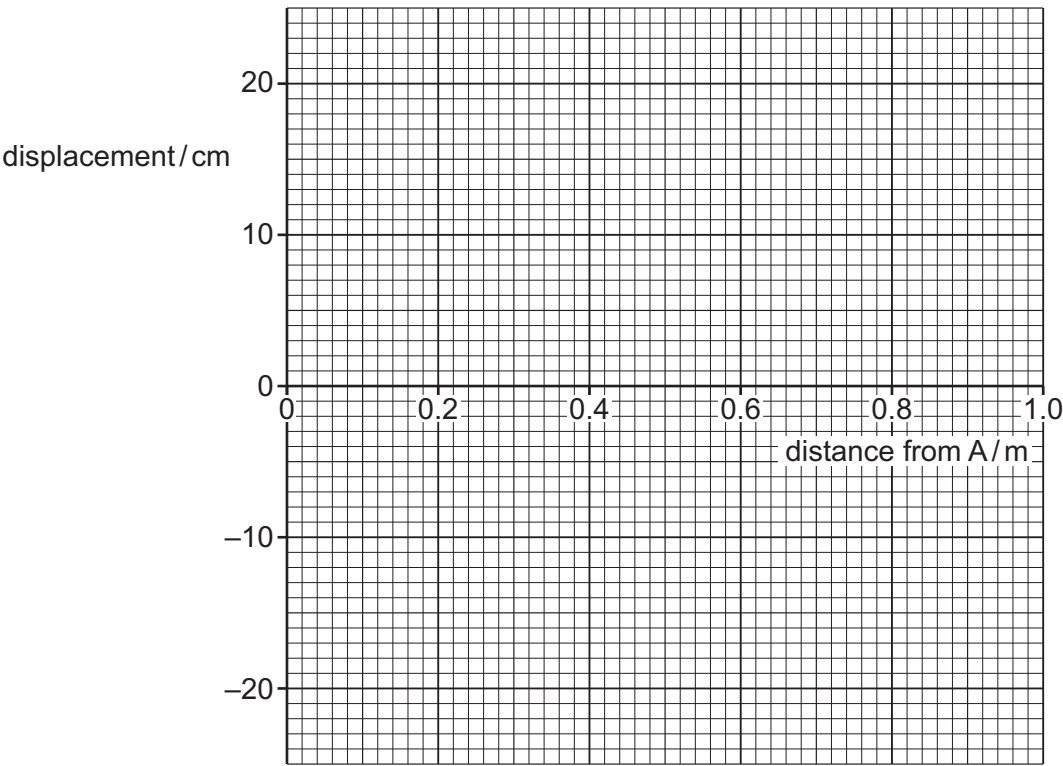


Fig. 3.2

- (d) The intensity of wave X is I_X . The intensity of wave Y is I_Y . Use Fig. 3.1 to determine the ratio $\frac{I_X}{I_Y}$. [2]